0.1 Randomization Inference for the Reduced-Form Interactions

We complement conventional clustered inference with randomization inference that reproduces the assignment of the four item arms. In each draw, we reallocate Control, Calendar, Ink, and Bracelet across communities within the six county by finalized-distance strata, preserving the realized number assigned to every arm in every stratum. Outcomes and all distance measures remain fixed. The exercise therefore uses only the randomized item assignment; it does not treat realized distance as if it had itself been permuted.

Table 1 presents three specifications. The first, and primary, specification interacts item assignment with the originally assigned Close/Far condition. The second uses the final realized Close/Far classification, and the third uses continuous realized centroid-to-PoT distance. These latter two specifications are sensitivity analyses: their randomization-based content comes from item assignment conditional on the displayed distance measure.

Table 1: Randomization inference for treatment-by-distance effects

Outcome	Contrast	Estimate	95% CI	Fisher p	Stud. RI p	RW p	N
Original assigned Close/Far							
Administrative take-up	Any Signal - No Signal	0.124	[0.047, 0.200]	0.006	0.003	0.006	9,805
Administrative take-up	Bracelet - Calendar	0.075	[-0.021, 0.171]	0.237	0.145	0.145	9,805
Definite peer classification	Any Signal - No Signal	0.136	[0.023, 0.248]	0.020	0.025	0.041	1,141
Definite peer classification	Bracelet - Calendar	0.242	[0.115, 0.370]	0.003	0.001	0.003	1,141
Correct classification	Any Signal - No Signal	0.124	[0.036, 0.211]	0.008	0.008	0.028	1,134
Correct classification	Bracelet - Calendar	0.156	[0.037, 0.275]	0.017	0.014	0.038	1,134
Perceived own observability	Any Signal - No Signal	0.120	[-0.025, 0.265]	0.111	0.121	0.121	1,141
Perceived own observability	Bracelet - Calendar	0.287	[0.108, 0.467]	0.005	0.004	0.014	1,141
Final realized Close/Far							
Administrative take-up	Any Signal - No Signal	0.068	[-0.014, 0.150]	0.134	0.118	0.249	9,805
Administrative take-up	Bracelet - Calendar	0.077	[-0.025, 0.180]	0.221	0.158	0.249	9,805
Definite peer classification	Any Signal - No Signal	0.132	[0.027, 0.237]	0.023	0.019	0.121	1,141
Definite peer classification	Bracelet - Calendar	0.148	[0.016, 0.279]	0.068	0.038	0.156	1,141
Correct classification	Any Signal - No Signal	0.100	[0.014, 0.187]	0.031	0.029	0.156	1,134
Correct classification	Bracelet - Calendar	0.076	[-0.042, 0.194]	0.242	0.225	0.455	1,134
Perceived own observability	Any Signal - No Signal	0.085	[-0.057, 0.227]	0.256	0.262	0.455	1,141
Perceived own observability	Bracelet - Calendar	0.123	[-0.062, 0.308]	0.240	0.218	0.455	1,141
Realized distance (per km)							
Administrative take-up	Any Signal - No Signal	0.052	[-0.003, 0.107]	0.096	0.085	0.201	9,805
Administrative take-up	Bracelet - Calendar	0.036	[-0.046, 0.118]	0.411	0.422	0.422	9,805
Definite peer classification	Any Signal - No Signal	0.119	[0.040, 0.197]	0.007	0.006	0.047	1,141
Definite peer classification	Bracelet - Calendar	0.138	[0.041, 0.235]	0.027	0.012	0.066	1,141
Correct classification	Any Signal - No Signal	0.077	[0.013, 0.141]	0.024	0.026	0.139	1,134
Correct classification	Bracelet - Calendar	0.073	[-0.025, 0.171]	0.123	0.168	0.416	1,134
Perceived own observability	Any Signal - No Signal	0.095	[-0.008, 0.198]	0.079	0.089	0.270	1,141
Perceived own observability	Bracelet - Calendar	0.165	[0.022, 0.308]	0.028	0.039	0.156	1,141

Notes: Any Signal is the average of Ink and Bracelet; No Signal is the average of Control and Calendar. Binary-distance estimates are differences-in-differences; continuous-distance estimates are differences in gradients per kilometer. All regressions adjust for gender, age, expected distance, and county fixed effects. The 95 percent intervals use community- clustered CR1 standard errors. Fisher columns compare the unstudentized statistic with its sharp-null randomization distribution. Studentized RI recomputes the clustered standard error under every allocation. RW denotes Romano–Wolf stepdown adjustment. Primary original-assignment and secondary realized-distance hypotheses are adjusted in separate, prespecified outcome families. All tests are two-sided.

The Fisher calculation is finite-sample exact for the sharp null that item assignment has no effect on any outcome. The studentized calculation targets the weaker null of a zero average interaction and is asymptotically justified; it is not described as an exact weak-null test. Observability estimates are conditional on respondents with a usable Table-A peer-knowledge module. We therefore report the available sample for each measurement outcome and interpret this exercise together with the appendix's selection and Lee-bound analyses.

Figure 1 displays the underlying community means. It makes clear both the number of independent assignment units and the distinction between the randomized binary assignment and realized continuous distance.

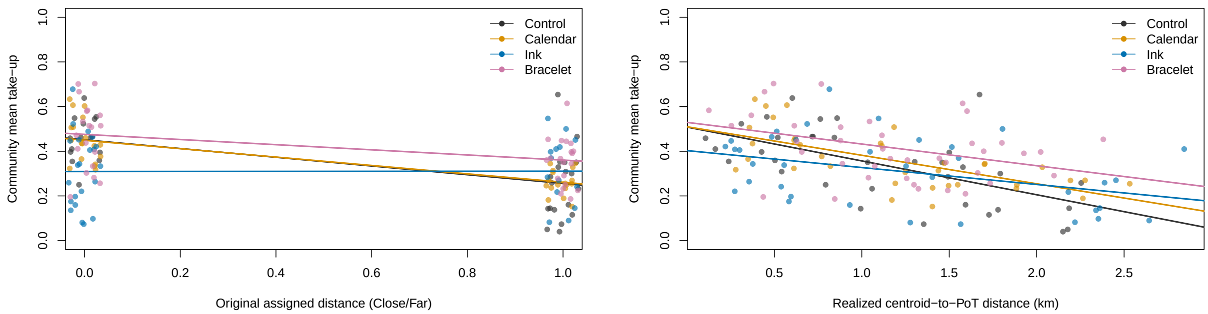

Figure 1: Community-level take-up by assigned and realized distance